Topic : Autonomous Crop Irrigation System using Artificial Intelligence

Savita Choudhary, Vipul Gaurav, Abhijeet Singh, and Susmit Agarwal (2019), in their paper titled "Autonomous Crop Irrigation System using Artificial Intelligence," published a paper in which they combined multiple machine learning techniques for rainfall and water-requirement prediction — using Random Forest for rainfall classification/regression, ARIMA for rainfall time-series forecasting, and PLSR (Partial Least Squares Regression) for estimating crop water requirements, along with expert irrigation decisions.

On the hardware part, a Raspberry Pi combined with NodeMCU / IoT modules was used to control the irrigation system.

The dataset was sourced from Government of India rainfall records, with the initial deployment focused on Punjab region, specifically wheat crops, along with soil and weather inputs.

For evaluation, the model used a 70/15/15 train/validation/test split, measuring both rainfall prediction accuracy and irrigation decision accuracy. In controlled-environment testing, the system achieved an overall accuracy of 80%, while for Punjab's annual rainfall prediction, the model predicted 545 mm against an actual 549.5 mm, giving an accuracy of 83%. Based on these predictions along with real-time soil moisture data, the pump was automatically controlled without manual input.

However, the system's major limitation is its dependency on external weather-station data, and the authors suggest replacing this with on-premise sensors for better rural deployment, additionally, the model's generalization to other crops and regions requires further adaptation.

The key strength of this work lies in its strong autonomy , the system independently predicts rainfall and water needs and decides pump ON/OFF status without requiring human intervention.